Retail Performance Dashboard · Aurora Living

Power BI portfolio project · Sushmita De

1. Project at a glance

Aurora Living is a fictional Indian home and lifestyle retailer with 20 stores, a website and an app across 5 regions, operating during 2023-2025.

The leadership wants one place to answer following questions:
Are we hitting the target?
Where do we lose profit?
Which customers matter most?
Are deliveries and marketing spends working?
I built a Power BI report that answers these from 3 years of sales, customer, delivery and marketing data.

Source Data	12 raw CSV files (120k orders, ₹1.35bn of net sales) simulated using python, including tables for orders, order lines, shipment details, product and customer details, etc.
What I built	Power Query cleaning, a snowflake data model (4 fact tables, 8 dimensions), 40+ DAX measures, 5 report pages plus a drill-through page, and dynamic row-level security
Skills used	Power BI · Power Query · DAX · data modelling · row-level security · data storytelling

2. Preparing the Data

The raw files came with realistic problems. I fixed them all in Power Query, so every fix is a recorded step that reruns on each refresh.

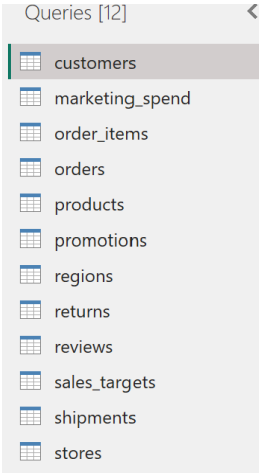
Problem in the raw data	How I fixed it
Duplicate orders (1.5%)	Remove Duplicates, so sales and orders aren't double-counted
Mixed date formats (3% written dd/mm/yyyy)	Change Type with locale English (India), which parses both formats in one step
Prices stored as text, e.g. “₹2,939.00”	Removed ₹ and commas, then converted to decimal
Negative quantities (0.3%)	Took the absolute value and flagged the row in a qty_corrected column, so the decision stays visible
One value spelt many ways: “north”, “NORTH ”, “North Region”; “Swift Ship”, “SWIFTSHIP”	Trim, Capitalize Each Word and Replace Values, so slicers and joins see one value
Blank promotion, return reason and sub-category	A “No Promotion” key (PRM00), “Not Specified” and “Unclassified”, so rows stay visible instead of appearing as (Blank)

Shaping the tables

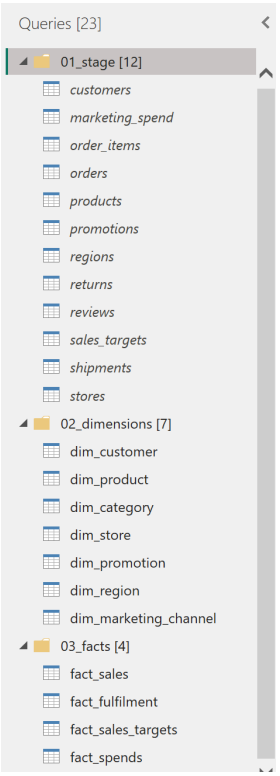
- **Merged files that share a grain:** order lines + returns became fact_sales; orders + shipments + reviews became fact_fulfilment.
- **Allocated shipping cost** from each order to its lines by value proportion, so product-level profit is correct.
- **Built customer lifetime metrics** (first and last order, order count, total spend) that drive the RFM segments and cohorts.
- **Maintained intermediate queries as staging** with load disabled, so the model holds only the final tables.

Before and after cleaning

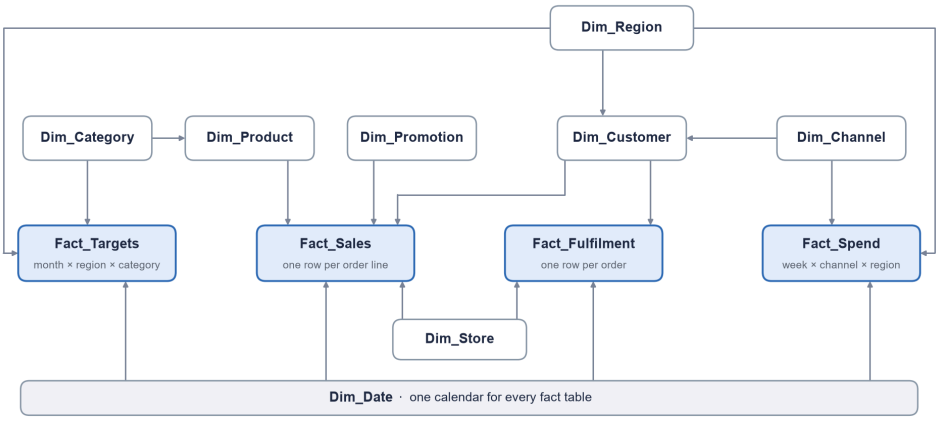
Before



After



3. Data model and DAX

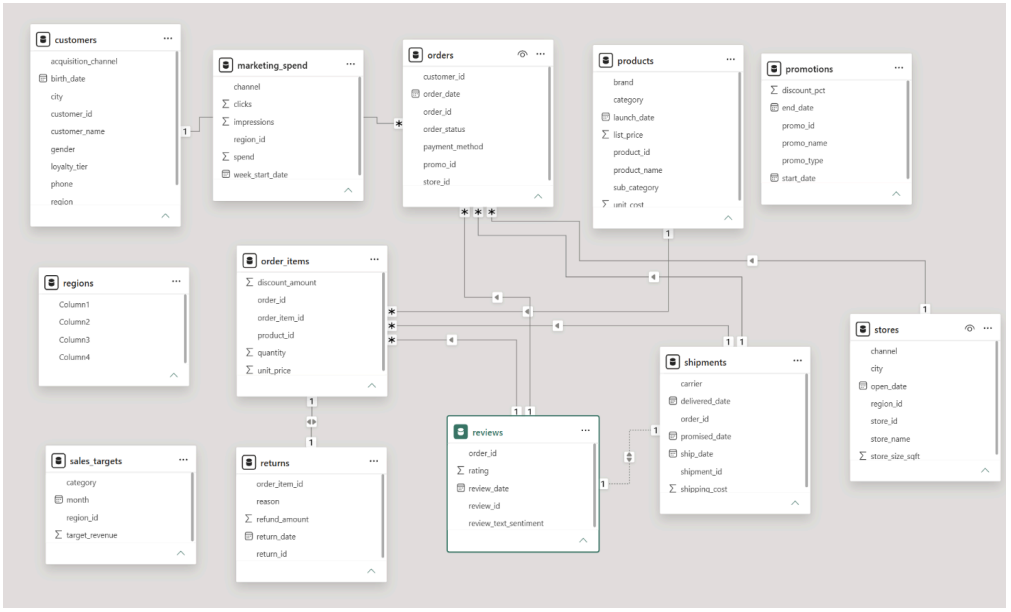


Snowflake schema: 4 fact tables at their own grain, joined only through shared dimensions. Arrows show filter direction.

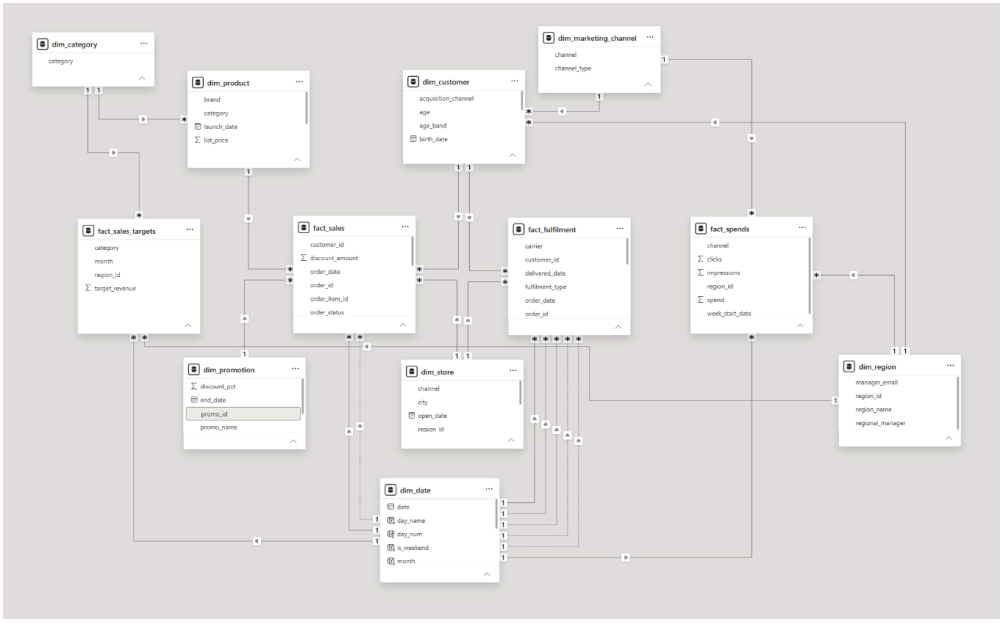
- **Region comes from the customer, not the store**, because online orders have no store region. One security rule on dim_region then filters all 4 fact tables.
- **Targets keep their own grain** (month x region x category), so fact_targets links to Date, Region and Category, not to products.
- **No fact-to-fact joins**, so filters never take an ambiguous path. Return, delivery and review dates use inactive relationships, switched on with USERELATIONSHIP.

Before and after modelling

Before



After

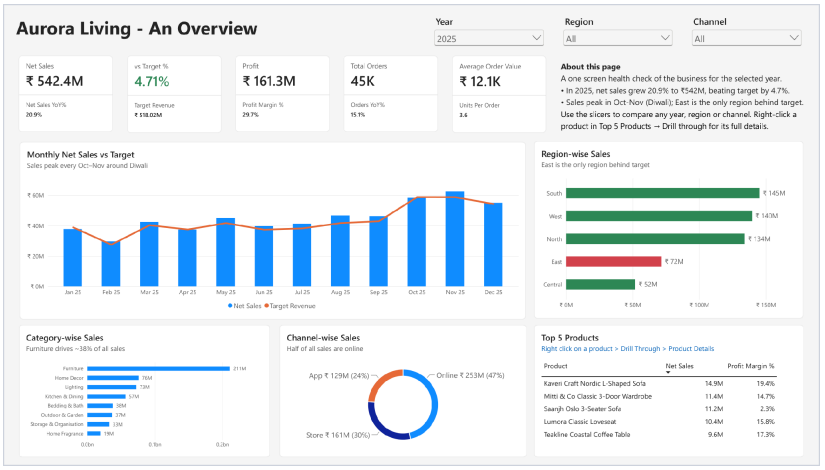


DAX highlights

Measure	How it works
Target Revenue	TREATAS maps monthly targets onto the calendar; the measure returns blank under store, product or promotion filters a target can't respond to.
YoY % and Net Sales LY	SAMEPERIODLASTYEAR, shown only when one year is selected, so multi-year views never show a misleading comparison.
Repeat rate, segments, retention	Context transition over customers, ALLSELECTED shares of total, and retention by quarters since first purchase.
Row-level security	[manager_email] = USERPRINCIPALNAME() on Dim_Region, so each regional manager sees only their region (tested with View as).

4. The dashboard, page by page

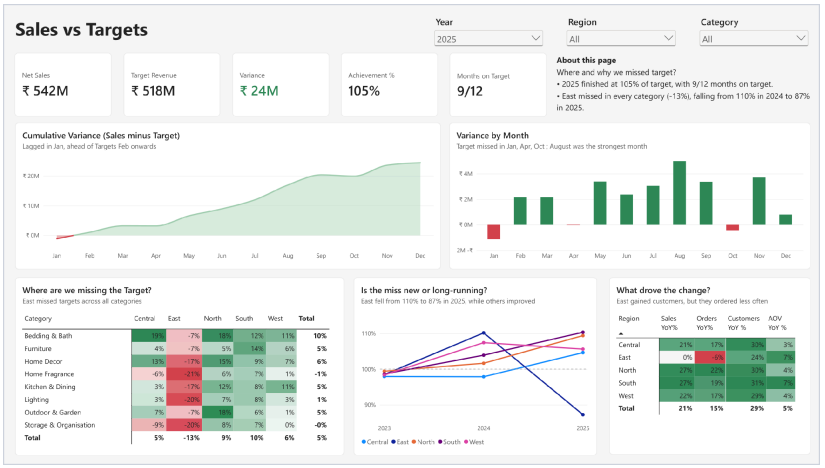
Each page answers one business question. On the 5 analysis pages, every chart has a subtitle that states its finding, an “About this page” box sums up the page, and Year and Region slicers let a manager focus on any year or region.



Executive Overview

Answers: How is the business doing, and is it on target?

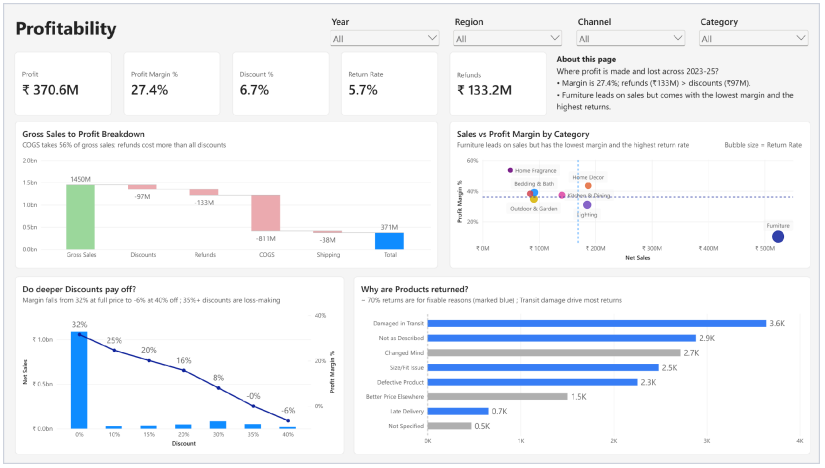
Shows: KPI cards with year-on-year change, monthly sales vs target, region bars coloured by target status, category and channel mix, and a Top 5 products table that drills through to product detail.



Sales vs Targets

Answers: Where did we miss the target, and why?

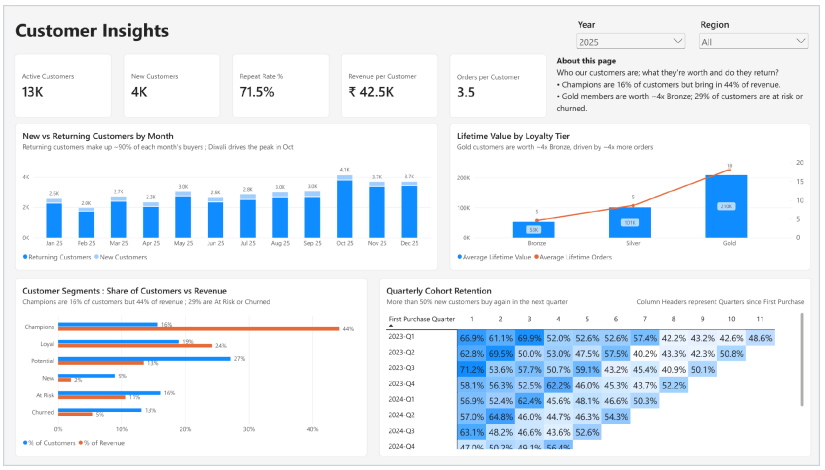
Shows: Cumulative and monthly variance, a category x region heatmap, a 3-year achievement trend by region, and a table that splits growth into orders, customers and order value.



Profitability

Answers: Where is profit made and lost?

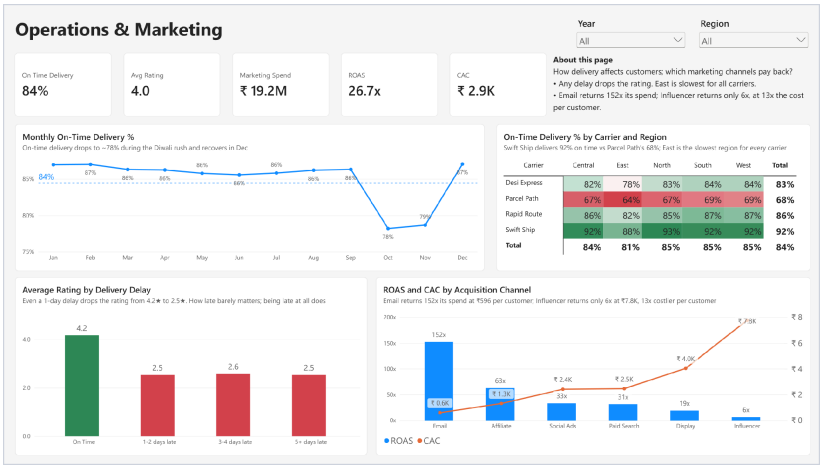
Shows: A gross-sales-to-profit waterfall, sales vs margin by category (bubble size = return rate), margin at each discount level, and return reasons with fixable causes in blue.



Customer Insights

Answers: Who are our best customers, and do they come back?

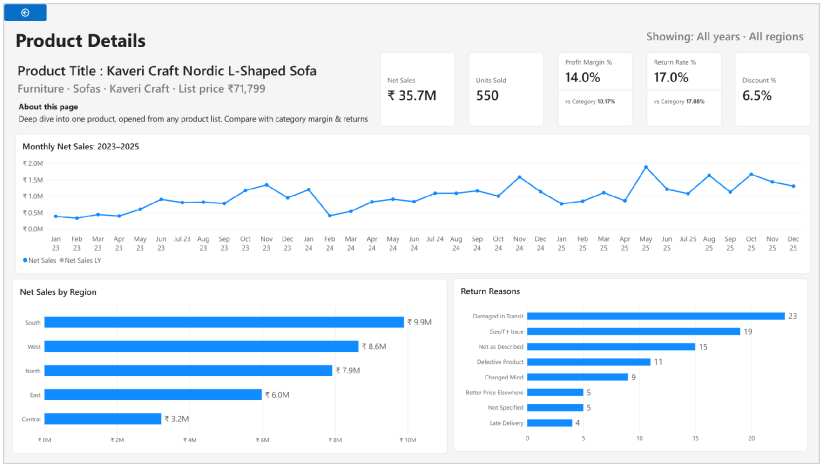
Shows: New vs returning customers by month, lifetime value by loyalty tier, customer segments by share of customers vs revenue, and a quarterly cohort-retention heatmap.



Operations & Marketing

Answers: Are we delivering on time, and is marketing paying back?

Shows: Monthly on-time delivery, on-time % by carrier and region, average rating by delivery delay, and ROAS vs CAC by acquisition channel.



Product Details (drill-through)

Answers: How is one product doing against its category?

Shows: Opened by right-clicking any product: a dynamic title and filter label, margin and return rate against the category average, a monthly trend that adds last year when one year is selected, sales by region and return reasons.

5. What the analysis found

Six findings, each with the action it supports. Sales figures are for 2025; the rest cover 2023-25.

Insights	Recommended actions
Growth is strong, but the East is slipping. 2025 net sales grew 20.9% to ₹542M, 4.7% above target. East fell from 110% of target in 2024 to 87%: it gained 24% more customers, but orders fell 6% as they bought less often.	Rebuild repeat buying in East with re-engagement campaigns and loyalty offers

Deep discounts lose money. Margin falls from 32% at full price to -6% at 40% off; discounts of 35% or more are loss-making.	Cap promotions below 35% off
Returns cost more than discounts. Refunds (₹133M) exceed all discounts (₹97M), and 70% of returns have fixable causes, led by transit damage. Furniture sells the most but has the lowest margin (10%) and the highest return rate (18%).	Improve packaging and product descriptions, starting with Furniture
Late deliveries hurt ratings. Any delay drops the average rating from 4.2 to about 2.5. Parcel Path is on time 68% of the time vs Swift Ship's 92%, and on-time delivery dips to 78% in the Diwali rush.	Shift volume to Swift Ship and add carrier capacity before October
A few customers drive most revenue. Champions are 16% of customers but 44% of revenue, while 29% of customers are At Risk or Churned.	Protect Champions with loyalty perks; run a win-back campaign for At Risk customers
Marketing efficiency varies 25x. Email returns 152x its spend at ₹596 per new customer; Influencer returns 6x at ₹7.8K per customer.	Shift budget from Influencer to Email and Affiliate